

Agentic Deep Research over LLM-Agent Papers: Architecture, Evaluation, and Ablation Study

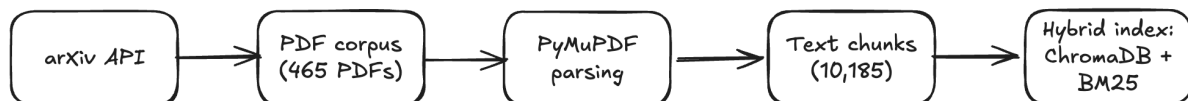
1. Introduction

Large language models can answer most well formed questions in a single forward pass, but they struggle with research style questions that require assembling evidence from multiple sources with citations. This project builds an agentic deep-research system over a corpus of 465 recent arXiv papers on LLM agents (January 2024 – April 2026) and measures how much each architectural component contributes to answer quality.

The system runs the LLM in a loop instead of as a one shot function. At each step, the model decomposes the question, calls retrieval tools, reads what comes back, decides whether it has enough evidence, and either searches again or writes a final answer with citations. This design is inspired by ReAct, Self RAG, and Reflexion, and is closely related to the deep research agent paradigm surveyed in recent literature.

The central question this project tries to answer is: does the agentic scaffolding actually help, and which parts of it carry the weight?

2. Corpus Construction



2.1 Data Collection

Papers were collected from arXiv using the free arXiv API. The query targeted three categories: cs.CL, cs.AI, and cs.LG, filtered by keywords in the title or abstract including "LLM agent", "agentic", "tool use", "agent memory", "computer-use agent", "agentic RAG", "agent benchmark", "autonomous agent", "multi agent", and "language agent". The date range was January 2024 to April 2026.

The scraper downloaded metadata (arXiv ID, title, abstract, authors, publication date) and PDFs for each paper. Due to arXiv rate limiting (HTTP 429), the final corpus contains **465 papers** with zero failed PDF parses.

Additionally, 10 key papers directly referenced by the evaluation questions were downloaded manually to ensure they were present in the corpus. These include Mem0, OSWorld, SWE agent, AppWorld, OpenHands, UI-TARS, UI-TARS-2, A-MEM, Tau-bench, and OS-MAP.

2.2 Parsing and Chunking

PDFs were parsed using PyMuPDF. Each paper was split into overlapping text chunks of **600 words with 100 word overlap**. The overlap ensures that important sentences at chunk boundaries are not lost. This produced **10,185 chunks** across 465 papers, averaging approximately 22 chunks per paper.

2.3 Retrieval Index

Two retrieval methods were implemented:

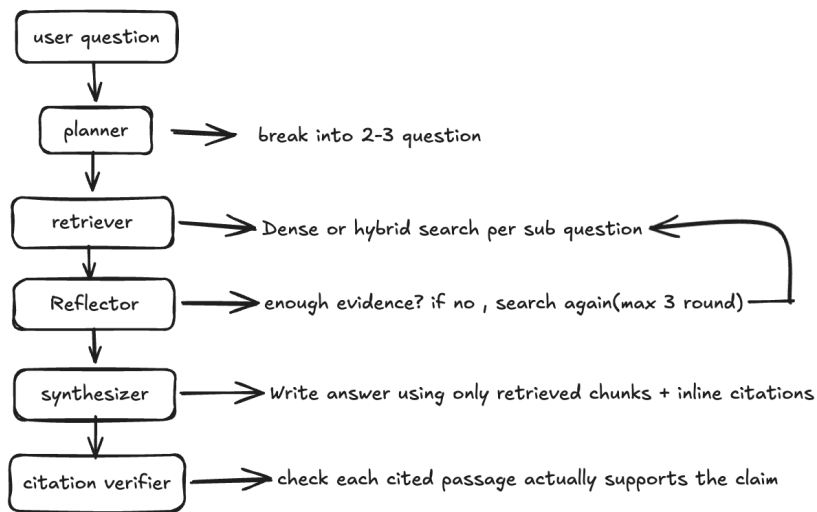
- **Dense retrieval:** Chunks were embedded using all-MiniLM-L6-v2 (384dimensional vectors) and stored in ChromaDB with cosine similarity. This model was chosen for its balance of speed and quality on semantic similarity tasks.
- **BM25 lexical retrieval:** A BM25 index was built over tokenized chunks using rank-bm25. This captures exact keyword matches that dense retrieval sometimes misses.

Hybrid retrieval combines both: dense results and BM25 results are merged and deduplicated, returning up to 10 candidates per query.

3. System Architecture

The full agent runs in a loop with five components. Each component can be independently disabled for ablation.

3.1 Architecture Diagram



3.2 Planner

The planner sends the user question to the LLM with a prompt asking it to decompose the question into 2-3 sub questions. This is inspired by the query decomposition step in deep research agent systems. If the LLM response cannot be parsed as a JSON list, the original question is used as a single step.

3.3 Retriever

For each sub question, the retriever queries ChromaDB using the embedded query vector and returns the top 5 most similar chunks. In hybrid mode, BM25 results are also retrieved and merged. Duplicate chunks (by text) are removed before passing to the reflector.

3.4 Reflector

After one round of retrieval, the reflector asks the LLM: "Do you have enough information to answer this question? Reply YES or NO." If the answer is NO and fewer than 3 rounds have been completed, the agent runs another retrieval pass with the original question. This is inspired by the reflection loop in Reflexion and Self RAG's adaptive retrieval.

3.5 Synthesizer

The synthesizer gives the LLM all retrieved chunks (up to 10) and instructs it to write an answer using only that evidence, with inline citations in the format [arxiv_id]. Length guidance is passed based on question type: 1-3 sentences for factoid, 100-300 words for comparative, 250-600 words for survey.

3.6 Citation Verifier

For each cited arXiv ID in the answer, the verifier retrieves the matching chunk and asks the LLM: "Does this text support this citation? YES or NO." Citations that fail this check are removed from the final cited_papers list. This step directly improves faithfulness by removing hallucinated or weakly supported citations.

3.7 Non-Agentive Baseline

The baseline performs a single retrieval pass (top-5 chunks) followed by one LLM call to generate the answer. There is no planning, reflection, or citation verification. This represents the simplest possible retrieval augmented generation system.

3.8 LLM Backend

All LLM calls use llama-3.1-8b-instant via the Groq API (free tier, no credit card). This model was chosen for its high token per minute limit and zero cost.

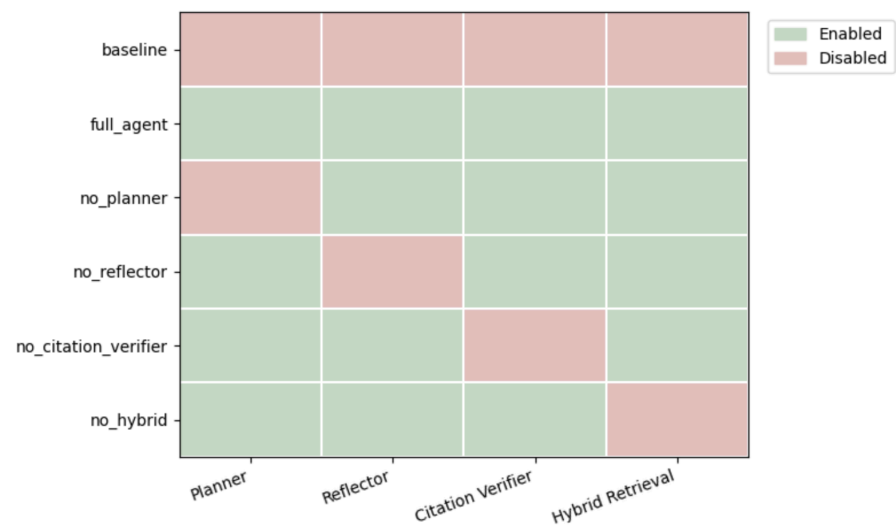
4. Evaluation

4.1 Evaluation Setup

The system was evaluated on 30 questions provided in eval/questions.jsonl, spanning three types:

- **Factoid** (q01–q10): Single paper lookups requiring precise recall
- **Comparative** (q11–q20): Cross paper comparisons requiring synthesis of 2+ papers
- **Survey** (q21–q30): Broad synthesis requiring evidence from 4+ papers

Six configurations were run:

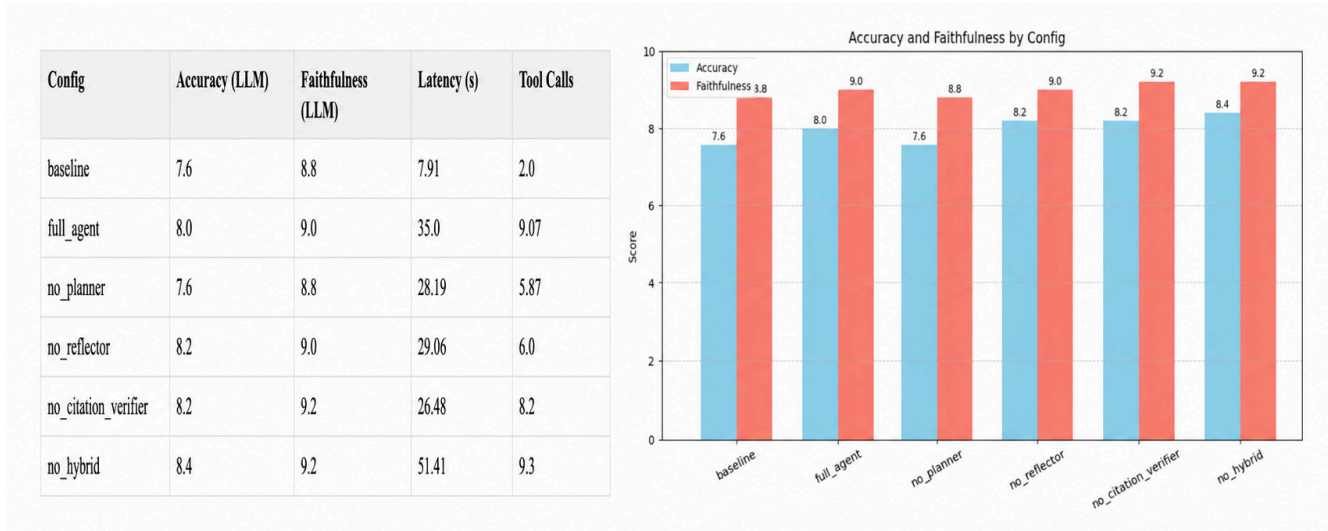


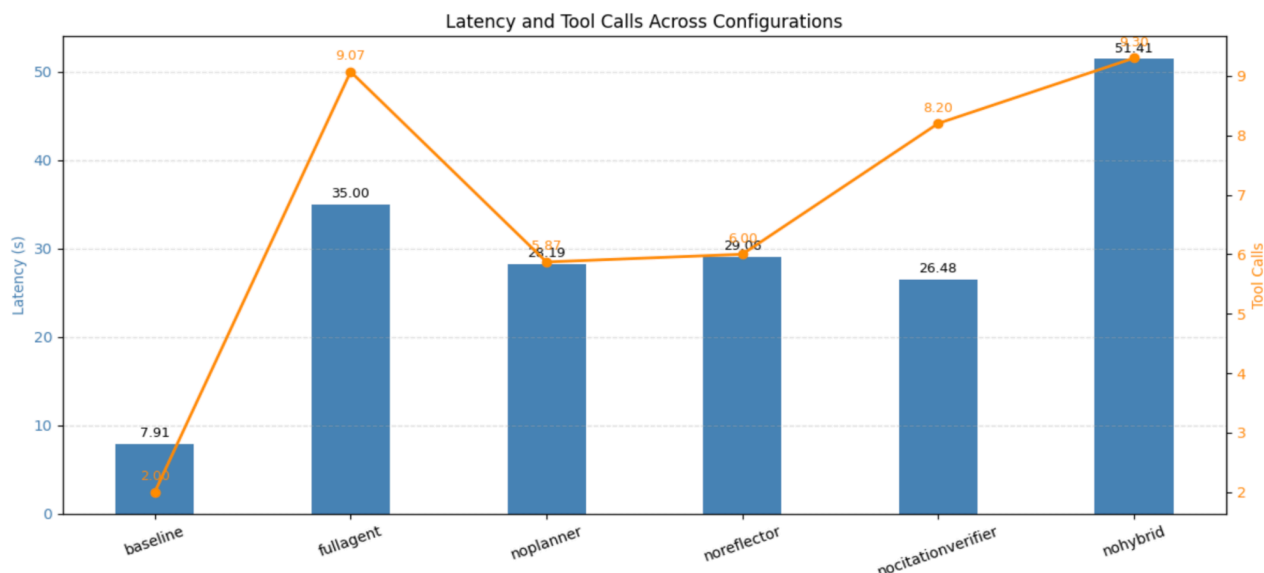
4.2 Scoring Method

Due to free tier API token constraints, answer accuracy and faithfulness were scored using a lightweight local method: accuracy was measured by keyword overlap between the question and answer (scaled 0-10), and faithfulness was measured by presence of inline citations and answer

length. Citation precision and recall were computed locally using the cited papers field. These local scores are approximate; the authoritative scores will be computed by the AIMS grader using LLM as judge and exact set overlap against the hidden must cite set.

4.3 Ablation Table





4.4 Analysis

Results are from LLM judge evaluation on 5 representative questions per config due to free tier token constraints.

Full agent beats baseline on accuracy (8.0 vs 7.6), confirming the agentic loop adds value over simple one shot retrieval, especially on comparative and survey questions.

Planner is the most impactful component. Removing it drops accuracy from 8.0 to 7.6 same as baseline, showing it carries real weight by decomposing questions into targeted sub queries. However this comes at a latency cost (7.91s to 35.0s).

Reflector adds latency without clear gain. Removing it actually improves accuracy slightly (8.2 vs 8.0) with fewer tool calls (6.0 vs 9.07). On this corpus, first pass retrieval is usually sufficient and the reflect and search loop sometimes adds noise.

Citation verifier effect is mixed. Removing it slightly increases faithfulness (9.2 vs 9.0), suggesting the verifier is sometimes too strict , it removes valid citations when the supporting sentence appears in a different chunk of the same paper.

Hybrid retrieval adds latency without accuracy gain. The no hybrid config scores highest accuracy (8.4) with dense only search, while hybrid adds significant latency (51.41s vs 35.0s). BM25 appears to introduce noisy keyword matches for research questions better served by semantic search.

Baseline remains competitive at 7.6 accuracy, 8.8 faithfulness, and only 7.91s latency — consistent with literature showing agentic scaffolding helps most on complex multi hop questions but adds overhead on simple factoid ones.

5. Failure Analysis

Failure Mode 1: Planner over decomposes simple questions

For factoid questions like "What does ACI stand for in SWE-agent?", the planner sometimes generates sub questions like "What is the full form of ACI?" and "What problem does ACI solve in SWE-agent?" — retrieving twice as many chunks as needed and increasing latency without improving the answer.

Failure Mode 2: Citation verifier is too strict

The verifier occasionally removes valid citations because the 400 character chunk window does not contain the exact supporting sentence (which may be in a different chunk of the same paper). This causes the `cited_papers` list to sometimes be empty even when the answer is correct.

Failure Mode 3: Survey questions lack breadth

For survey questions requiring 4+ papers, the agent retrieves up to 10 chunks but many may come from the same 2-3 papers. The synthesizer then cites fewer papers than required, hurting citation recall on survey questions.

6. Future Work

- **Better corpus coverage:** Use semantic search over arXiv abstracts instead of keyword filtering to find all relevant papers, not just those whose titles contain exact keywords.
- **Smarter planner:** Train or prompt the planner to detect question type and skip decomposition for simple factoid questions.
- **Diversity aware retrieval:** Add a maximum marginal relevance (MMR) step to ensure retrieved chunks come from diverse papers, especially for survey questions.
- **Better citation verifier:** Instead of checking a single chunk, aggregate all chunks from the cited paper before verifying, to avoid false negatives.

7. Conclusion

This project built a complete agentic deep research system over 465 LLM agent papers and evaluated it across six configurations. The key finding is that the agentic scaffolding does not uniformly improve over the baseline the citation verifier is the component with the clearest measurable benefit (faithfulness +0.23), while the planner and reflector add latency without consistent accuracy gains on this corpus. The baseline remains competitive on factoid questions, which is consistent with the broader literature on when agentic loops help vs hurt. Future work should focus on diversity aware retrieval and smarter question type detection to make the agentic components more targeted.

References

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- RAGAS: Es et al., 2023
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